

```

1 import components.standard.Standard;
2
3 /**
4  * Interface for a Pomodoro Timer kernel component with primary methods.
5  *
6  * This interface defines methods to manage a Pomodoro like timer with the basic
7  * functions of setting, starting, pausing, stopping, and resting.
8  *
9  * @initially <pre>
10 *   ensures
11 *     isRunning = false and
12 *     remainingTime = 0
13 * </pre>
14 */
15 public interface TimerKernel extends Standard<Timer> {
16
17     /**
18      * Sets the timer for the given number of minutes provided by user.
19      *
20      * @param minutes
21      *     the number of minutes to set for the timer
22      * @updates remainingTime
23      * @ensures remainingTime = minutes * 60
24      */
25     void setTimer(int minutes);
26
27     /**
28      * Starts the countdown of the timer.
29      *
30      * @updates remainingTime
31      * @ensures Timer counts down from remainingTime to 0 unless paused or
32      *     stopped.
33      */
34     void startTimer();
35
36     /**
37      * Pauses the current timer countdown.
38      *
39      * @updates isRunning
40      * @ensures isRunning = false and remainingTime is unchanged
41      */
42     void pauseTimer();
43
44     /**
45      * Stops the timer and resets the time.
46      *
47      * @updates isRunning, remainingTime
48      * @ensures isRunning = false and remainingTime = 0
49      */
50     void stopTimer();
51
52     /**
53      * Sets and starts a rest timer after a work session.
54      *
55      * @param minutes
56      *     number of minutes to rest
57      * @updates remainingTime
58      * @ensures Timer is reset to minutes * 60 and starts counting down
59      */
60     void restTime(int minutes);
61

```

```
62T    /**
63     * Returns the current remaining time in minutes and seconds.
64     *
65     * @return the number of seconds remaining in the timer
66     */
67    int getRemainingTime();
68
69    /**
70     * Returns whether the timer is currently running.
71     *
72     * @return true if the timer is running, false otherwise
73     */
74    boolean isRunning();
75 }
76
```